AI Speech Intelligence & Ward Survey Architecture

Project: Old Bowenpally (Ward 119), Hyderabad — Multilingual Household Speech & Electoral Intelligence

Focus: Speech-to-Text (Telugu-Hindi-English code-switching), Gemini Structured Extraction, 12-Sheet Relational System, and Automated Executive Reporting.

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1. End-to-End System Pipeline

The system converts raw, consented field interview audio recorded on Android mobile devices into structured ward analytics and automated daily briefings:

1. **Field Recording:** Mobile devices (ODK Collect / KoBoToolbox) record short 30–90 second household interviews with GPS and consent.
2. **Multilingual Speech Recognition (ASR):** Audio is processed via Groq Whisper or Sarvam AI, biased with a local Ward 119 Gazetteer to prevent spelling drift on proper nouns.
3. **Gemini Structured Extraction:** Gemini 2.5 Flash normalizes colloquial dialect, preserves proper nouns verbatim, and outputs strict JSON matching the analytical framework.
4. **Central Database & 12-Sheet Export:** Extracted data is ingested into PostgreSQL and exported into a 12-sheet relational Excel workbook.
5. **Automated Reporting:** Google Looker Studio or Python Jinja scripts automatically generate daily 1-page executive briefs for the Corporator and campaign war room.

2. The Master 12-Sheet Architecture

#	Sheet Name	On-Site Operational Purpose	Key Columns & Data Fields
1	**Dashboard**	One-screen executive command center with live KPI summaries and priority alerts.	Top Red-Zone Colonies, High-Urgency Issues, Overdue Civic Actions, Coverage Gaps.
2	**Ward Master**	Official baseline reference list for all dropdowns to prevent spelling errors.	Polling Booth No., Colony Name, Boundary/Section, Key Landmarks, Registered Voters.
3	**Codebook**	Standardized definitions and allowed values across all sheets.	Issue Category, Urgency Scale (1–5), Recurrence (Chronic/Monsoon/ New), Scope (House/Street/Colo

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4	**Survey Coverage**	Field audit sheet tracking sampling targets and identifying surveyor bias.	Booth No., Target Households, Actual Surveyed, Age Mix, Gender Split, Community Split.
5	**Issue Log**	Core master log: one row per grievance with verbatim evidence and audio link.	Submission ID, Colony, Issue Type, Root Cause, Impacted Group, Urgency, Verbatim Telugu Quote, Audio ID.
6	**Action Tracker**	Municipal accountability tracker for government engineers and supervisors.	Grievance ID, Proposed Fix, Department (GHMC/HMWSSB), Assigned Official (AE/Inspector),

#	Sheet Name	On-Site Operational Purpose	Key Columns & Data Fields
			Deadline, Status.
7	**Promise Tracker**	Tracks campaign commitments made by the candidate vs. real delivery.	Promise Description, Location (e.g., Dhobi Ghat Park), Target Date, Current Delivery Status.
8	**Stakeholder Directory**	Registry of local influencers, caste heads, and community leaders.	Influencer Name, Colony, Phone Number, Group Represented, Vote Base Swayed, Current Stance.
9	**Cadre Deployment**	Management sheet for party booth workers and local coordinators.	Booth No., In-Charge Name, Assigned Lanes/Streets, Daily Household Targets,

#	Sheet Name	On-Site Operational Purpose	Key Columns & Data Fields
			Oversight Notes.
10	**Opposition Log**	Counter-intelligence tracking rival party movements and ground rumors.	Colony/Booth, Rival Party Active, Promises Made, Local Rumors, Risk Level, Counter-Message.
11	**Welfare Desk**	Citizen-specific casework for individual government schemes.	Citizen Name, House No., Contact, Scheme (Aasara Pension/Ration/2BHK), Issue Detail, Status.
12	**Outreach Calendar**	Schedule of upcoming campaign events, camps, and allocated funds.	Event Type (Padayatra/Help Camp/WhatsApp Push), Date, Target Colony, Budget Allocated, Reach.

3. Ground Implementation: Ward 119 Specifics

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  Red-Zone Civic Hotspots: Anjaiah Nagar (silt extortion by sanitation workers, Jamia Masjid drainage), Bhavani Nagar (severe monsoon inundation), Mallikarjuna Nagar (sewage mixing in Manjeera drinking water), Manovikas Nagar (unfenced open nala hazard).
- **Key Influencers to Mobilize:** Advocate Mohammed Idris (Manovikas Nagar — ~600 swing voters, Phone: 9866883786), Mohammed Khader, Dr. Janardhan, Basthi Youth Committees.
- **Coordinator Oversight:** Restraining abrasive local intermediaries (e.g., Irfan in Mala Basthi, Raju Yadav in Yadava Basthi) to prevent alienation of core voters.

4. Recommended Platform Stack

Stage	Recommended Platform	Why It Works Best
Audio Collection	ODK Collect / KoBoCollect (Android)	100% offline field operation, automated GPS tagging, audio compression.
Speech Transcription	Groq Whisper / Sarvam AI API	High-speed transcription with native Telugu & Indian English code-switching

Stage	Recommended Platform	Why It Works Best
		accuracy.
Extraction & Translation	Google Gemini 2.5 Flash API	Enforced JSON schemas, gazetteer entity preservation, low latency and cost.
Data Ingestion & Storage	Streamlit Web App + openpyxl (or PostgreSQL)	Single-file Python prototype, automatic multi-sheet Excel writing, zero frontend complexity.
Executive Reporting	Google Looker Studio (Automated Email)	Connects to Google Sheets/Excel; automatically emails daily PDF reports to leadership at 8:00 AM.